ACADEMIC PROJECT • PS-12

VehicleHistory

Blockchain-based ownership & service history



Sahas Bochare 08

Solidity • Sepolia • Ethers.js • MetaMask

The record breaks when the vehicle changes hands.

Fragmented ownership

Transfers live across paper files, dealer systems and informal records.

Unverifiable service claims

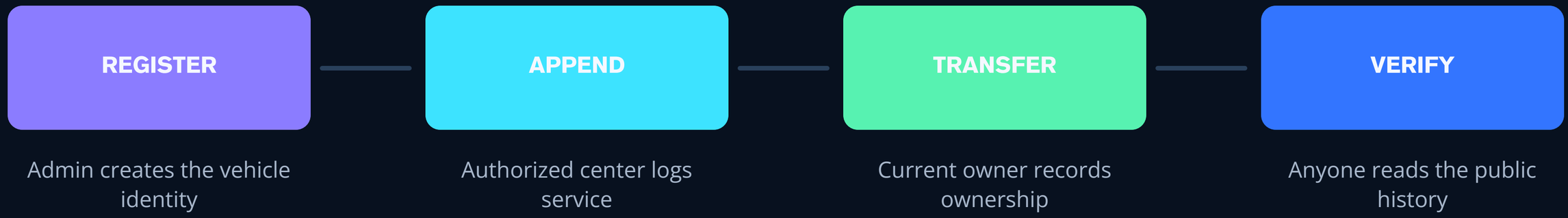
Records can be missing, altered or impossible to trace to an authorized source.

High-trust purchase

Buyers must make a major decision without one reliable timeline.



One vehicle. One tamper-evident timeline.



Shared truth replaces private paperwork.

What the solution must make possible

Immutable history

Append-only ownership and maintenance events

Role-based writes

Only approved actors can change records

Public verification

VIN lookup without privileged access

Traceable provenance

Every event carries sender and block time

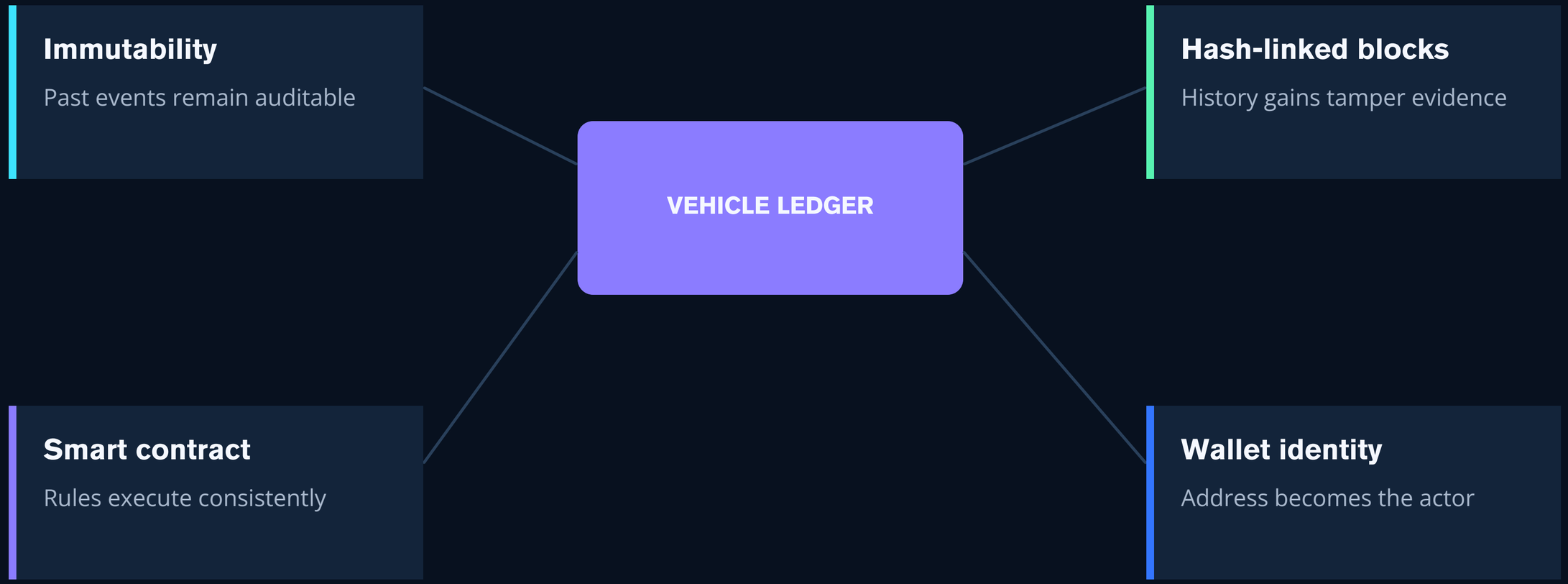
Wallet-native access

MetaMask signs every protected action

Demo-ready UX

Clear dashboard, status and QR entry point

Blockchain concepts, applied—not decorative



From user intent to on-chain proof



Reads are public. Protected writes require a signed transaction and contract authorization.

VehicleHistory.sol — the trust boundary

Core data

Vehicle
- VIN / metadata
- current owner
- registered flag

ServiceRecord
- date / mileage
- description / center

State-changing functions

registerVehicle()
addServiceRecord()
transferOwnership()
authorizeServiceCenter()

Public read functions

getVehicle()
getServiceHistory()
getCurrentOwner()
isAuthorizedCenter()

Mappings connect VIN → vehicle → ordered service records.

Four roles. Clear permissions.

ADMIN

Register vehicle
Authorize center

SERVICE CENTER

Append verified
maintenance

CURRENT OWNER

Transfer ownership
to next wallet

PUBLIC USER

Search VIN
Read full history

CREATE → MAINTAIN → TRANSFER → VERIFY

Demo: a complete vehicle story in four transactions



01 **Vehicle registered**

02 **Service center authorized**

03 **Service record appended**

04 **Ownership transferred**

✓ Each successful write emits an auditable on-chain event

Security is proven by what fails

VALID PATHS

PASS

Authorized admin registers vehicle

PASS

Approved center adds service

PASS

Owner transfers ownership

DELIBERATE FAILURE CASES

REVERT

Duplicate VIN registration

REVERT

Unapproved wallet adds service

REVERT

Non-owner attempts transfer

A public history page turns blockchain data into trust

Dashboard

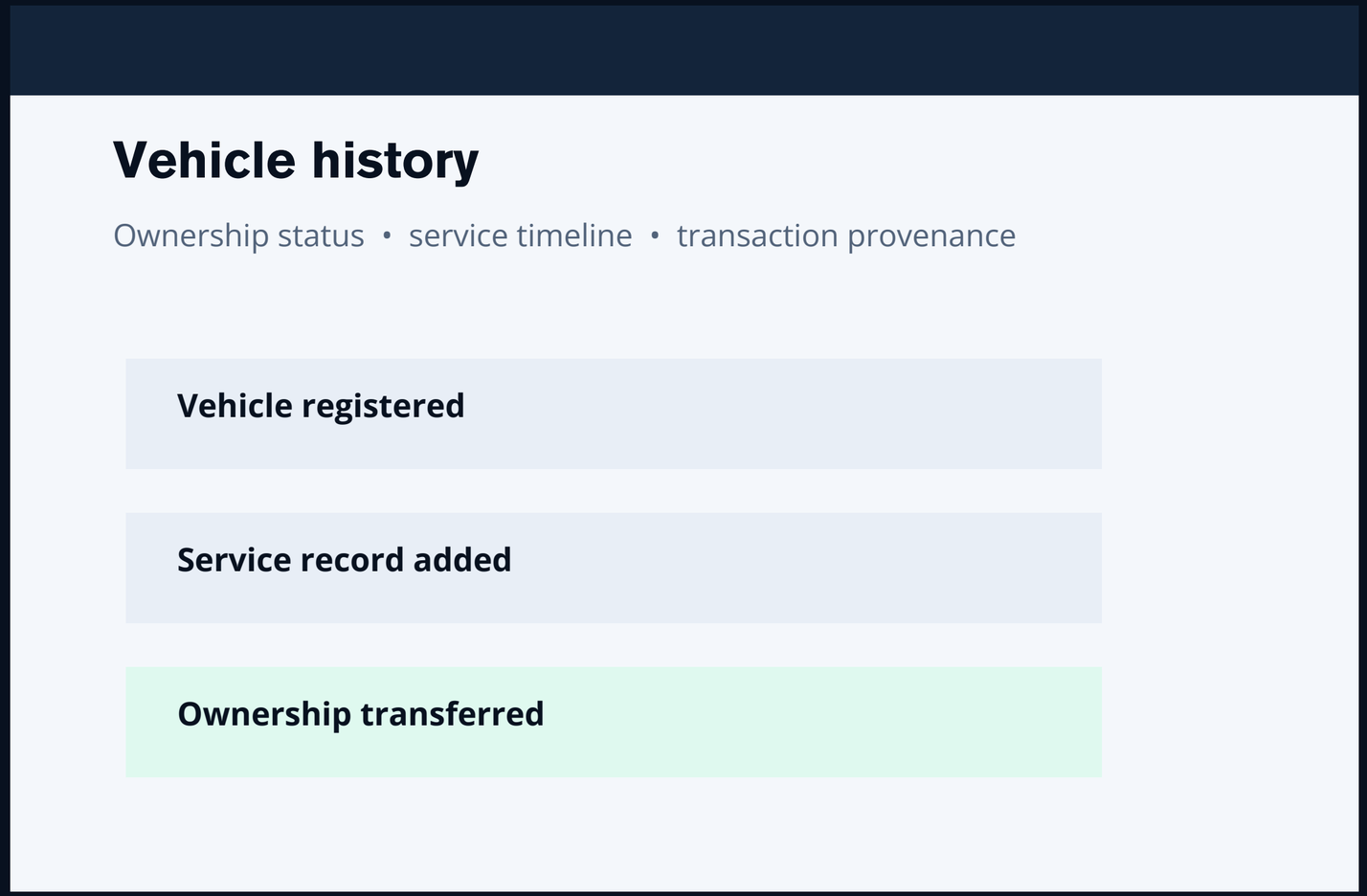
Connect wallet, search VIN and show role-aware actions.

Public history

Readable timeline sourced directly from contract data.

QR concept

A vehicle-linked QR opens the public VIN route instantly.



Vehicle history becomes shared infrastructure.

VERIFY, DON’T ASSUME.

VehicleHistory demonstrates how smart contracts can preserve provenance, enforce write permissions and give buyers a transparent timeline.

01 **Deploy to a production-ready network**

02 **Add document hashes and decentralized storage**

03 **Integrate trusted workshops and transport authorities**

04 **Improve privacy with selective disclosure**